MILITARY SPECIFICATION

OBTURATOR
M753 8 INCH PROJECTILE

This specification is approved for use by the US Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers requirements, quality assurance provisions and the preparation for delivery criteria for obturators used in M753, 8 Inch Projectiles (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-B-121	Barrier Material, Greaseproof, Waterproof, Flexible
MIL-A-48078	Ammunition, Standard Quality Assurance Provisions, General Specification for

Beneficial comments (recommendations, additions, deletions), and any pertinent data which may be of use in improving this document, should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAN-Q, Dover, New Jersey 07801, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426), appearing at the end of this document, or by letter.

FSC 1110

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:22Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-1169	Packaging, Packing and Marking for Shipment of Inert Ammunition Components

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS (see 6.6)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9326205	Obturator

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Construction. The obturator shall comply with all requirements specified on drawing (dwg.) 9326205, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the dwg. and herein.

3.2 Material. The contractor shall provide objective evidence (see 6.4) that materials comply with applicable dwgs. and specifications.

3.3 Breaking load. The obturator shall have a minimum breaking load of 400 lbs. and a minimum elongation of ten percent. Testing shall be performed in accordance with 4.5.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:22Z
Check the source to verify that this is the current version before use.

3.4 First article. This specification contains technical provisions for first article inspection (see 4.3). Requirements for submission of first article samples shall be as specified in the contract.

3.5 Workmanship. Obturators shall be fabricated in a thoroughly workmanlike manner and there shall be no evidence of porosity, inclusions, ripples, voids, surface imperfections or any other obvious plastic defects. Finished parts shall be clean and free of dirt, grease, dust and any other foreign matter. Cleaning methods shall not be injurious to the parts nor shall the parts be contaminated by the cleaning agents.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract or purchase order, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspection. The inspections specified herein are classified as follows:

1. First Article Inspection (see 4.3)
2. Quality Conformance Inspection (see 4.4)

4.3 First article inspection.

4.3.1 Submission. When the contract requires first article inspection (see 3.4), the contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:22Z
Check the source to verify that this is the current version before use.

Part Description	Drawing	Quantity
Obturator	9326205	20

4.3.2 Inspections to be performed. The first article sample submitted in accordance with 4.3.1 shall be subjected to any or all of the requirements of the applicable dwg. and to the tests and inspections specified in the first article inspection table herein. All parts shall have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract and all materials shall be obtained from the same source of supply as will be used in regular production.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:22Z
Check the source to verify that this is the current version before use.

FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-O-63170(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.3.2.1	Obturator	1 OF 1	9326205
			NEXT HIGHER ASSEMBLY: M753 Proj.

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Classification of Defects and Tests	20			4.4.2.1

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:22Z
Check the source to verify that this is the current version before use.

4.3.3 Inspection equipment. The inspections and tests specified in 4.3.2 shall be accomplished with the gages and test equipment specified in 4.4.4.

4.3.4 Rejection. See MIL-A-48078.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, an inspection lot of parts shall not exceed one days production and shall be produced from a single lot of approved material (see 3.2).

4.4.2 Examinations. See MIL-A-48078 and the quality conformance inspection tables herein. Unless otherwise specified, sample destructive tests shall be performed in accordance with Table I.

Table I. Sampling size for destructive tests.

LQ = .10 Acceptance No. = 0 Consumers Risk = .10

No req'd to be delivered	Sample size for no failures
11-18	16
19-25	17
26-37	18
38-57	19
58-105	20
106-249	21
250 and over	22

Lot size required = No required to be delivered + Sample size

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:22Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-O-63170(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Obturator (see note (c))	1 OF 1	9326205
			NEXT HIGHER ASSEMBLY: M753 Proj.

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major:					
101.	Inside diameter	(a)		3.1	Gage/TMC
102.	Width between smallest inside diameter and largest outside diameter	(a)		3.1	Gage/TMC
103.	Height of step between smallest inside diameter and largest inside diameter	(a)		3.1	Gage/TMC
104.	Width of smallest inside diameter	(a)		3.1	Gage/TMC
105.	Overall width	(a)		3.1	Gage/TMC
106.	Width between smallest inside diameter and smallest outside diameter of beveled surface	(a)		3.1	Gage/TMC
107.	Bevel Angle	(a)		3.1	Gage/TMC
108.	Breaking load and elongation	(b)		3.3	4.5.1
109.	Marking (identification dots)		100%	3.1	Visual
110.	Poor workmanship		100%	3.5	Visual
111.	Material (dwg 9287987)		---	3.2	Visual

NOTES:
(a) See 4.4.3.1, (b) See Table I, (c) Dimensions apply after conditioning at 70 F and 50% R.H. for 24 hrs min. per dwg. 9326205

Source: https://assist.dla.mil - 7 Downloaded: 2026-10-23T02:22Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET		DRAWING NUMBER
4.4.2.2	Packaging		1 OF 1		See 5.0
					NEXT HIGHER ASSEMBLY: N.A.
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major:					
101.	Packaging and packing not in accordance with applicable requirements		---	5.0	Visual
102.	Objective evidence (see 3.2) not available for packaging and packing material		---	5.1.1	Visual
103.	Parts not clean and free of dirt and other foreign material prior to packaging		100%	5.1.1	Visual
104.	Parts not individually packaged		100%	5.1.1	Visual
105.	Stiffening and cushioning of packages inadequate to provide firm package (when required)		100%	5.1.1	Visual/Manual
106.	Individual packages not properly heat sealed		100%	5.1.1	Visual
107.	Intermediate pack improper or not properly sealed		100%	5.1.2	Visual/Manual
108.	Exterior pack improper or not properly sealed		100%	5.1.3	Visual
109.	Marking of individual packages, intermediate packs or exterior packs missing, improper or illegible		100%	5.1.4	Visual
110.	Evidence of poor workmanship		100%	-----	Visual

NOTES:

Source: https://assist.daps.dla.mil -- Downloaded: 2026-10-23T02:22Z
Check the source to verify that this is the current version before use.

4.4.3 Testing. Testing is described in the first article and quality conformance inspection tables and herein.

4.4.3.1 Tool manufacturing control. When tool manufacturing control (TMC) is specified, the first ten pieces initially produced from each mold cavity inspected for the applicable classification of defects herein shall constitute initial mold acceptance. Subsequently, the first and last ten pieces from each mold cavity from each days production inspected for the applicable classification of defects herein shall constitute lot acceptance. In addition, any time the mold requires repair an additional ten pieces from each mold cavity shall be inspected for the applicable classification of defects herein before resumption of production. If any sample fails to meet any of the applicable inspection requirements, the lot of production under consideration shall be rejected.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract (see section 6 of MIL-A-48078 and 6.3 herein).

4.5 Test methods.

4.5.1 Breaking load (see 3.3). The obturator shall be tested for breaking load and elongation using approved test equipment and fixtures. The obturator shall be assembled on the test fixture and stretched at a speed of 2.00 inches per minute until the obturator breaks with the elongation and load recorded. Measurement of elongation shall not commence until both sides of the obturator are stretched parallel by the approved test fixture. If the obturator fails to meet the breaking load and elongation requirements of 3.3, it shall be rejected.

5. PACKAGING

5.1 Packaging and packing. Unless otherwise specified in the contract or order, packaging and packing shall be in accordance with MIL-STD-1169 subject to the following details and conditions.

5.1.1 Packaging. Items shall be single item packaged in fiberboard containers, sealed in waterproof barrier bags and heat sealed in accordance with MIL-STD-1169, Level A. Barrier bags shall be fabricated from

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:22Z
Check the source to verify that this is the current version before use.

MIL-B-121 (Type II, Grade A, Class 1) material or an approved equivalent. Fillers, positioners, stiffeners and cushioning adequate to prevent physical damage to the item and to finished surfaces during transit, storage and handling shall be provided as required.

5.1.2 Intermediate packing. Single item packages shall be packed in intermediate packages in accordance with MIL-STD-1169, Level B.

5.1.3 Packing. Intermediate packages shall be packed in accordance with MIL-STD-1169, Level B.

5.1.4 Marking. Unit packages, intermediate packs and exterior containers shall be marked with the following:

a. Item nomenclature
b. Part or dwg. no.
c. Used on M753 Projectile
d. Manufacturers
e. Quantity enclosed
f. Lot no.
g. Contract or purchase order no.
h. Date of acceptance

6. NOTES

6.1 Intended use. Obturators covered by this specification are intended for use in motor body bonding assemblies for rocket motors for M753, 8 Inch Projectiles.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and data of this specification.
b. First article inspection, when required (see 3.4).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD1423, Contract Data Requirements List (CDRL) the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:22Z
Check the source to verify that this is the current version before use.

contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph no.	Data requirements	Applicable DID no.
4.4.4	Inspection equipment approval	DI-R-1714 (Tailored)

(Copies of DID's required by contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 Equipment design approval. Submit equipment designs, as required, to Commander, U.S. Army Armament Research and Development Command, ATTN: DRDAR-QAN-I, Dover, New Jersey 07801.

6.4 Objective evidence for material. A statement (certification) supported by test data that each lot of material meets the requirements of the applicable dwgs. and specifications.

6.5 Supersession data. This specification includes the requirements of AR-PD-26 dated 9 April 1979 with Amendmend 1 dated 24 April 1981.

6.6 Drawings and specifications. Drawings and specifications listed in Section 2 of this specification under the heading U.S. Army Armament Research and Development Command may also include dwgs. and specifications prepared and identified as Frankford Arsenal, Edgewood Arsenal, Rock Island Arsenal or Picatinny Arsenal dwgs. and specifications. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

Custodian

Army-AR

Preparing activity

Army-AR

(Project 1110-A163)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:22Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

BUSINESS REPLY CARD

FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Armament R&D Command
ATTN: DRDAR-QAN-Q
Dover, NJ 07801

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:22Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-O-63170 OBTURATOR M753 8 INCH PROJECTILE

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426

EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:22Z
Check the source to verify that this is the current version before use.